

BCSE0004: CRYPTOGRAPHY & NETWORK SECURITY

Objective: This Course focuses towards the introduction of network security using various cryptographic algorithms and understanding network security applications and practical applications that have been implemented and are in use to provide email and web security.

Credits:03

L-T-P-J:3-0-0-0

Module No.	Content	Teaching Hours
I	Introduction: Introduction to Security Attacks, Services and Mechanism, Classical Encryption Techniques-Substitution Ciphers and Transposition Ciphers, Steganography, Stream and Block Ciphers, Cryptanalysis. Symmetric Key Cryptosystems: Block Cipher Principles, Shannon's Theory of Confusion and Diffusion, Data Encryption Standard(DES), Strength of DES, Triple DES, Advance Encryption Standard (AES), Linear and Differential Cryptanalysis, Block Ciphers Modes of Operation. Introduction to Number Theory: Modular Arithmetic, Prime and Relative Prime Numbers, Primitive Roots, Fermat's and Euler's Theorem, Extended Euclidean Algorithm, Chinese Remainder Theorem. Algebraic Structures: Introduction to Group, Ring & Field of the Form GF(P). Asymmetric Key Cryptosystems: RSA Cryptosystem, Attacks on RSA, Security of RSA, Discrete Logarithm Problem, Elgamal Encryption Algorithm.	20
II	Hash Functions and Macs: Authentication Functions, Message Authentication Code, Hash Functions, Birthday Attacks, Security of Hash Functions, Secure Hash Algorithm [SHA-512]. Digital Signature: Digital Signatures, RSA Digital Signature Scheme, Elgamal Digital Signature Techniques, Digital Signature Standards(DSS). Key Management: Symmetric Key Distribution, Diffie Hellman Key Exchange Algorithm. Public Key Distribution: X.509 Certificates, Public Key Infrastructure. Authentication Applications and E-Mail Security: Kerberos, Pretty Good Privacy (PGP), S/MIME. IP Security and Web Security: IP Sec Architecture, Authentication Header, Encapsulating Security Payloads, Combining Security Associations, Key Management, Introduction to Secure Socket Layer, Transport Layer Security, Secure Electronic Transaction (SET). System Security: Introductory Idea of Intrusion, Intrusion Detection, Malicious Programs, Firewalls.	22

Text Books:

- W. Stallings , "Cryptography and Network Security: Principles and Practices", 8th Edition, Pearson Education, 2010.

Reference Books:

- B. A. Forouzan, "Cryptography & Network Security", 3rd Edition, Tata McGraw Hill, 2003.
- Wenbo Mao, "Modern Cryptography: Theory and Practice", Prentice Hall, 2003.
- Douglas Stinson, "Cryptography Theory and Practice", 2nd Edition, Chapman & Hall/CRC.

Outcome: After completion of course, the student will be able to:

- CO1: Understands the basic concepts of cryptography.
- CO2: Apply the symmetric key concepts of DES and AES for securing data.
- CO3: Apply the concepts of number theory of Asymmetric key cryptosystem.
- CO4: Understand the concepts of hash function, MAC and digital signature for data integrity.
- CO5: Explain the symmetric and asymmetric key distribution techniques.
- CO6: Understand the concepts of security mechanism at TCP/IP layer.